

Determinants

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Learning Outcomes

At the end of this lecture, you should be able to:

- Know what is the determinant of a matrix and its properties.
- Understand the Cofactor Expansion Theorem and know how to use it to find the determinant of a matrix.
- Know what happens to the determinant when we perform various operations on rows or columns and able to use these to efficiently calculate the determinants.
- Determine if a matrix is invertible by its determinant.

Determinants

Let $A = [a_{ij}] \in M_n(\mathbb{R})$. The **determinant** of A , denoted by $\det A$ or $|A|$ is a scalar defined by

$$\det A = \begin{cases} a_{11} & \text{if } n = 1, \\ a_{11}A_{11} + a_{12}A_{12} + \cdots + a_{1n}A_{1n} & \text{if } n \geq 2, \end{cases}$$

where, for each $1 \leq i, j \leq n$, let $A_{ij} = (-1)^{i+j} \det M_{ij}$ and $M_{ij} \in M_{n-1}(\mathbb{R})$ is an $(n-1)$ -submatrix of A obtained by deleting the i^{th} row and the j^{th} column of A .

The number A_{ij} is called the **cofactor** of a_{ij} in $\det A$ and $\det M_{ij}$ is called the **minor** of a_{ij} in $\det A$

This way of evaluating $\det A$ is called **cofactor expansion**.

Determinants

Example:

1. Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$.

2. $\det \begin{bmatrix} -3 & -2 & 4 \\ 4 & 3 & 1 \\ 0 & 2 & 4 \end{bmatrix}$

Determinants

Show that $\det \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} = aei + bfg + cdh - afh - bdi - ceg.$

Determinants

$$\det \begin{bmatrix} 2 & -1 & 0 & 0 \\ 3 & -3 & 2 & -2 \\ 4 & 2 & 0 & 0 \\ 2 & 0 & 0 & -1 \end{bmatrix}.$$

$$-2 \begin{vmatrix} 2 & -1 & 0 \\ 4 & 2 & 0 \\ 2 & 0 & -1 \end{vmatrix}$$

$$-2(-1)(4 + 4)$$

$$= 16$$

Cofactor Expansion Theorem

The determinant of an $n \times n$ matrix A can be computed using the cofactor expansion along any row or any column of A .

If $A = [a_{ij}] \in M_n(\mathbb{R})$, then

- (i) the expansion along i^{th} row $a_{i1}A_{i1} + a_{i2}A_{i2} + \cdots + a_{in}A_{in} = \det A$.
 - (ii) the expansion along j^{th} column $a_{1j}A_{1j} + a_{2j}A_{2j} + \cdots + a_{nj}A_{nj} = \det A$.
- For each $1 \leq i, j \leq n$, the number A_{ij} is called the cofactor of a_{ij} in $\det A$.

Example

By expanding the 3rd column, find

$$\det \begin{bmatrix} 2 & -1 & 0 & 0 \\ 3 & -3 & 2 & -2 \\ 4 & 2 & 0 & 0 \\ 2 & 0 & 0 & -1 \end{bmatrix}$$

What happens if you expand the cofactor along a different parallel row or column?

Cofactor Expansion Theorem

Let $A = [a_{ij}] \in M_n(\mathbb{R})$. If A is either a triangular or diagonal matrix, then the determinant of A is equal to the product of the diagonal entries of A ($\det A = a_{11}a_{22} \cdots a_{nn}$).

Example:

1.
$$\begin{vmatrix} 2 & 0 & 0 & 0 \\ 0 & -3 & 0 & 0 \\ 0 & 0 & 5 & 0 \\ 0 & 0 & 0 & -1 \end{vmatrix}$$

2.
$$\begin{vmatrix} -1 & 0 & 0 \\ 2 & 4 & 0 \\ 1 & 3 & 2 \end{vmatrix}$$

3.
$$\begin{vmatrix} -2 & -1 & 2 & 0 \\ 0 & -3 & 3 & -2 \\ 0 & 0 & 0 & 5 \\ 0 & 0 & 0 & -1 \end{vmatrix}$$

Properties of determinants

Let $A \in M_n(\mathbb{R})$. Then,

- $\det A = \det A^T$.
- If A contains a zero row or a zero column, then $\det A = 0$.
- If A contains two identical rows or two identical columns, then $\det A = 0$.

Example:

$$1. \begin{vmatrix} 0 & -1 & 2 & 0 \\ 2 & -3 & 3 & -2 \\ 0 & 0 & 4 & 1 \\ 0 & 0 & 2 & -1 \end{vmatrix} \qquad \begin{vmatrix} 0 & 2 & 0 & 0 \\ -1 & -3 & 0 & 0 \\ 2 & 3 & 4 & 2 \\ 0 & -2 & 1 & -1 \end{vmatrix}$$

$$2. \begin{vmatrix} 1 & 3 & -1 \\ 2 & 4 & 1 \\ 1 & 3 & -1 \end{vmatrix} \qquad \begin{vmatrix} 2 & 0 & 3 & 0 \\ -1 & -3 & 1 & -3 \\ 2 & 4 & 2 & 4 \\ 0 & 5 & 3 & 5 \end{vmatrix}$$

Properties of determinants

Let $A, B \in M_n(\mathbb{R})$. Then,

- If B is obtained from A by multiplying one row of A by a nonzero constant c , then $\det B = c \det A$.
- If B is obtained from A by interchanging two rows, then $\det B = -\det A$.
- If B is obtained from A by adding a multiple of one row of A to another row, then $\det B = \det A$.
- If $E \in M_n(\mathbb{R})$ is an elementary matrix, then $\det(EA) = \det E \det A$

Properties of determinants

So, we can use elementary row operations to transform a square matrix to a triangular (or diagonal) matrix, then we can compute the determinant more easily.

Example:

Find the determinant of the matrix $\begin{bmatrix} 3 & -1 & 1 & 1 \\ 3 & -1 & 2 & 1 \\ 0 & 2 & 4 & 3 \\ -3 & 1 & 1 & -2 \end{bmatrix}$.

Properties of determinants

Example:

Let A and B be square matrices such that B can be obtained from A by elementary row operations:

$$A \xrightarrow{R_1 \rightarrow R_1 + \frac{2}{9}R_2} A_1 \xrightarrow{R_2 \leftrightarrow R_3} A_2 \xrightarrow{R_2 \rightarrow 4R_2} B.$$

Find $\det A$ if $B = \begin{bmatrix} 3 & -1 & 1 & 1 \\ 3 & -1 & 2 & 1 \\ 0 & 2 & 4 & 3 \\ -3 & 1 & 1 & -2 \end{bmatrix}$.

Properties of determinants

Since $\det A = \det A^T$ for any n -square matrix A , then we can also determine the relationship between determinant and **elementary column operation**. So, if $A, B \in M_n(\mathbb{R})$. Then,

- If B is obtained from A by multiplying one column of A by a nonzero constant k , then $\det B = k \det A$.
- If B is obtained from A by interchanging two columns, then $\det B = -\det A$.
- If B is obtained from A by adding a multiple of one column of A to another column, then $\det B = \det A$.
- If $E \in M_n(\mathbb{R})$ is an elementary matrix, then $\det(AE) = \det A \det E$

Properties of determinants

Let $A \in M_n(\mathbb{R})$. Then A is invertible if and only if $\det A \neq 0$.

Example:

Determine if each of the following matrices is invertible.

$$A = \begin{bmatrix} 3 & -1 & 1 & 1 \\ 3 & -1 & 2 & 1 \\ 0 & 2 & 4 & 3 \\ -3 & 1 & 1 & -2 \end{bmatrix},$$

$$B = \begin{bmatrix} 1 & -3 & 1 \\ -2 & 6 & 1 \\ 1 & -3 & -1 \end{bmatrix}$$

Properties of determinants

Example:

For what values of a is $A = \begin{bmatrix} 1 & 2 & -1 \\ 3 & 1 & 0 \\ a & 0 & 1 \end{bmatrix}$ invertible?

Properties of determinants

Let $A, B \in M_n(\mathbb{R})$ and c be a scalar. Then

- $\det(cA) = c^n \det A$;
- $\det(AB) = \det A \det B$;
- if A is invertible, then $\det(A^{-1}) = \frac{1}{\det A}$.

Example:

Suppose that $A = \begin{bmatrix} -2 & 0 & 4 \\ 4 & -6 & -8 \\ -4 & -2 & 6 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 & 1 \\ 0 & -1 & 0 \\ 1 & 0 & 2 \end{bmatrix}$.

Properties of determinants

Example:

Suppose that A is a 4×4 matrix and $\det A = 3$. Find $\det B$ if B is given by

1. A^T .
2. A with second and last rows swapped.
3. A with first two rows swapped and then the last two column swapped.
4. A with the last column multiplied by 4.
5. $3A$.
6. A with the last row added by 5 times the first row.
7. A^{-1} .